Mikkel, Hannibal og Abdallah - Hand-in Assigment 3

Chit Chat service

Mikkel Vincent Hedegaard (mivh@itu.dk)
Hannibal Marcellus Munk (hmun@itu.dk)
Abdallah Jawher Alhamedi (aalh@itu.dk)

October 25, 2026 • Group : Mikkel, Hannibal og Abdallah • https://github.com/Honningbolden/mandatory-exercise-3

---

1 A) - discuss, whether you are going to use server-side streaming, client-side streaming, or bidirectional streaming?

We use server-side streaming to deliver messages to the clients. When a client joins, it opens a stream to the server, which stays open for as long as the client is connected. Through this stream, the client receives a continuous flow of broadcast made up of chat messages, join and leave notifications, each with its Lamport timestamp. This way, users can have full conversations on Chit Chat and are notified when other users join or leave. Sending a message works differently: a client publishes with a unary RPC, since this is an occasional, single request that does not need a persistent stream.

We chose server-side streaming because all messages pass through the server before reaching the other clients, so most of the traffic flows from the server to the clients. This is exactly the pattern server-side streaming is designed for. Client-side streaming would not work, since clients need to receive messages pushed by the server at any time. Bidirectional streaming would also work, but it adds complexity on both the client and the server without much benefit, because publishing is infrequent and fits naturally as a unary call.

2 B) - describe your system architecture - do you have a server-client architecture, peer-to-peer, or something else?

We have a server-client architecture, where a single server acts as the central hub and multiple clients connect to it. Clients do not communicate with each other directly; every interaction goes through the server. A client can join the chat, publish messages, and leave, and the server delivers all broadcasts to the connected clients.

The server keeps track of all currently active clients and their streams. The server keeps track of all clients through local channels stored in a map. When a client publishes a message, or when a client joins or leaves, the server updates its Lamport clock and broadcasts the message to every active client by sending it on each client's stream. Each client connection is served by its own goroutine, so a slow or disconnected client does not block delivery to the others. The server and each client run as separate processes and communicate only through gRPC.

3 C) describe what RPC methods are implemented, of what type, and what messages types are used for communication

We have implemented three message types: User, Message, and Nothing. User contains the username and a Lamport timestamp. Message has a variable sender as a string (username of sender), the content of the message being sent, and the Lamport timestamp. All updates to clients happen in the form of a Message. Nothing is a placeholder for an empty object. We have implemented three RPC methods.

1. The first one is PostMessage, which is unary. The client sends a `Message` to the server and receives `Nothing` once the chat message has been posted. The server then broadcasts the message to all connected clients.
2. The second one is Connect, which is server-side streaming. The client sends its user information as a `User` and, once connected, receives a stream of `Message`. The server creates a channel for the client and runs a loop that uses `select` to wait for messages on that channel, sending each one to the client through the stream. If the users channel is closed or the clients stream stops, the server removes the client from its list and the loop stops.
3. The third one is Disconnect, which is also unary. The client sends its user information as a `User` and receives `Nothing` once it has been disconnected. The server removes the client from its map and closes its channel, which ends the client's Connect stream. Our client does not currently use it, since closing the stream has the same effect, but it is available as an explicit way to leave.

4 D) - describe how you have implemented the calculation of the timestamps

Every Message and User also carries a Lamport timestamp. The server and each client keep their own internal integer clock, starting at 0. 1. Before a client sends a request (Connect or PostMessage) it increments its internal clock and attaches the value. 2. When the server receives a request it updates its internal clock as: clock = max(clock, received) + 1. Both the server and the client have a tick() helper function for this. A client leaving closes its stream. The server detects that and increments its own clock as a local event, since the client doesn't send a timestamp when cancelling the terminal session. 3. The server includes the updated value of its internal clock for each broadcast message, so all clients receive the same timestamp (logical time). When a client receives a broadcast it updates its own clock as: clock = max(clock, received) + 1.

5 D) - provide a diagram, that traces a sequence of RPC calls together with the Lamport timestamps

The diagram below describes the first few messages sent between Alice, Bob and Mikkel. The logs are in the assignment folder. C is the internal clock for each thread.

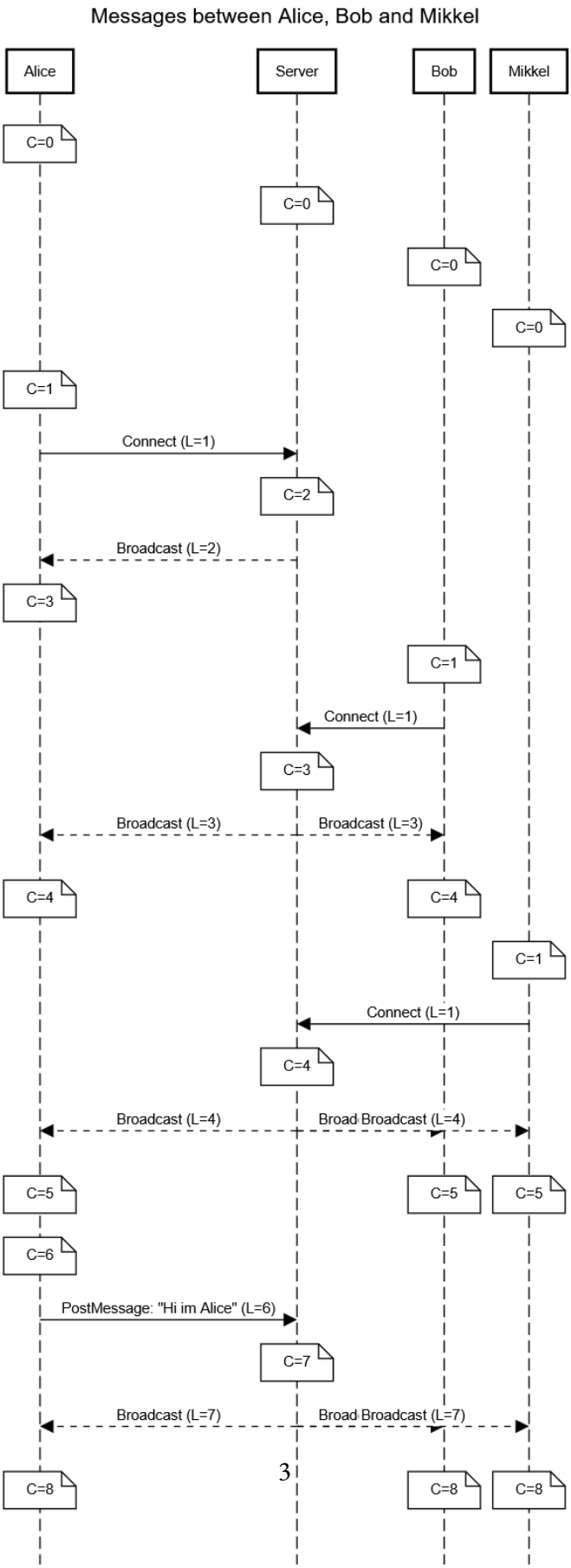


Figure 1: Conversation logs

6 E) Logs from our program

Server.log

```
[Server] 2026/10/24 23:43:19.018486 event=STARTUP addr=localhost:5050 lamport=0
[Server] 2026/10/24 23:43:28.738709 event=CONNECT client=alice lamport=2
[Server] 2026/10/24 23:43:28.738772 event=BROADCAST kind=JOIN client=alice lamport=2
[Server] 2026/10/24 23:43:28.738802 event=SEND to=alice from=Server lamport=2 msg="Participant alice joined Chit Chat at logical time
      2"
[Server] 2026/10/24 23:43:36.455280 event=CONNECT client=bob lamport=3
[Server] 2026/10/24 23:43:36.455330 event=BROADCAST kind=JOIN client=bob lamport=3
[Server] 2026/10/24 23:43:36.455347 event=SEND to=bob from=Server lamport=3 msg="Participant bob joined Chit Chat at logical time 3"
[Server] 2026/10/24 23:43:36.455347 event=SEND to=alice from=Server lamport=3 msg="Participant bob joined Chit Chat at logical time 3"
[Server] 2026/10/24 23:43:47.344503 event=CONNECT client=mikkel lamport=4
[Server] 2026/10/24 23:43:47.344563 event=BROADCAST kind=JOIN client=mikkel lamport=4
[Server] 2026/10/24 23:43:47.344582 event=SEND to=mikkel from=Server lamport=4 msg="Participant mikkel joined Chit Chat at logical
      time 4"
[Server] 2026/10/24 23:43:47.344581 event=SEND to=alice from=Server lamport=4 msg="Participant mikkel joined Chit Chat at logical time
      4"
[Server] 2026/10/24 23:43:47.344586 event=SEND to=bob from=Server lamport=4 msg="Participant mikkel joined Chit Chat at logical time 4"
[Server] 2026/10/24 23:44:02.104891 event=RECEIVE from=alice lamport=7 msg="Hello I'm alice!"
[Server] 2026/10/24 23:44:02.104963 event=BROADCAST kind=MESSAGE client=alice lamport=7
[Server] 2026/10/24 23:44:02.105022 event=SEND to=alice from=alice lamport=7 msg="Hello I'm alice!"
[Server] 2026/10/24 23:44:02.105032 event=SEND to=bob from=alice lamport=7 msg="Hello I'm alice!"
[Server] 2026/10/24 23:44:02.105037 event=SEND to=mikkel from=alice lamport=7 msg="Hello I'm alice!"
[Server] 2026/10/24 23:44:09.010358 event=RECEIVE from=bob lamport=10 msg="Hi! I'm bob"
[Server] 2026/10/24 23:44:09.010454 event=BROADCAST kind=MESSAGE client=bob lamport=10
[Server] 2026/10/24 23:44:09.010518 event=SEND to=alice from=bob lamport=10 msg="Hi! I'm bob"
[Server] 2026/10/24 23:44:09.010557 event=SEND to=bob from=bob lamport=10 msg="Hi! I'm bob"
[Server] 2026/10/24 23:44:09.010516 event=SEND to=mikkel from=bob lamport=10 msg="Hi! I'm bob"
[Server] 2026/10/24 23:44:12.874177 event=RECEIVE from=mikkel lamport=13 msg="I'm Mikkel"
[Server] 2026/10/24 23:44:12.874269 event=BROADCAST kind=MESSAGE client=mikkel lamport=13
[Server] 2026/10/24 23:44:12.874312 event=SEND to=mikkel from=mikkel lamport=13 msg="I'm Mikkel"
[Server] 2026/10/24 23:44:12.874340 event=SEND to=alice from=mikkel lamport=13 msg="I'm Mikkel"
[Server] 2026/10/24 23:44:12.874330 event=SEND to=bob from=mikkel lamport=13 msg="I'm Mikkel"
[Server] 2026/10/24 23:44:15.260879 event=RECEIVE from=mikkel lamport=16 msg="Mikkel!!"
[Server] 2026/10/24 23:44:15.260948 event=BROADCAST kind=MESSAGE client=mikkel lamport=16
[Server] 2026/10/24 23:44:15.260966 event=SEND to=alice from=mikkel lamport=16 msg="Mikkel!!"
[Server] 2026/10/24 23:44:15.260980 event=SEND to=bob from=mikkel lamport=16 msg="Mikkel!!"
[Server] 2026/10/24 23:44:15.260985 event=SEND to=mikkel from=mikkel lamport=16 msg="Mikkel!!"
[Server] 2026/10/24 23:44:27.296589 event=RECEIVE from=bob lamport=19 msg="Hello Mikkel. Omg it's sooo nice to meet youu"
[Server] 2026/10/24 23:44:27.296628 event=BROADCAST kind=MESSAGE client=bob lamport=19
[Server] 2026/10/24 23:44:27.296710 event=SEND to=alice from=bob lamport=19 msg="Hello Mikkel. Omg it's sooo nice to meet youu"
[Server] 2026/10/24 23:44:27.296656 event=SEND to=bob from=bob lamport=19 msg="Hello Mikkel. Omg it's sooo nice to meet youu"
[Server] 2026/10/24 23:44:27.296716 event=SEND to=mikkel from=bob lamport=19 msg="Hello Mikkel. Omg it's sooo nice to meet youu"
[Server] 2026/10/24 23:44:33.177027 event=RECEIVE from=alice lamport=22 msg="Tihi this is working very well"
[Server] 2026/10/24 23:44:33.177088 event=BROADCAST kind=MESSAGE client=alice lamport=22
[Server] 2026/10/24 23:44:33.177099 event=SEND to=alice from=alice lamport=22 msg="Tihi this is working very well"
[Server] 2026/10/24 23:44:33.177107 event=SEND to=mikkel from=alice lamport=22 msg="Tihi this is working very well"
[Server] 2026/10/24 23:44:33.177106 event=SEND to=bob from=alice lamport=22 msg="Tihi this is working very well"
[Server] 2026/10/24 23:44:35.318824 event=RECEIVE from=alice lamport=25 msg="Very cool"
[Server] 2026/10/24 23:44:35.318877 event=BROADCAST kind=MESSAGE client=alice lamport=25
[Server] 2026/10/24 23:44:35.318904 event=SEND to=mikkel from=alice lamport=25 msg="Very cool"
[Server] 2026/10/24 23:44:35.318925 event=SEND to=bob from=alice lamport=25 msg="Very cool"
[Server] 2026/10/24 23:44:35.318932 event=SEND to=alice from=alice lamport=25 msg="Very cool"
[Server] 2026/10/24 23:44:40.615790 event=RECEIVE from=bob lamport=28 msg="Indeed. Tres cool"
[Server] 2026/10/24 23:44:40.615817 event=BROADCAST kind=MESSAGE client=bob lamport=28
[Server] 2026/10/24 23:44:40.615833 event=SEND to=alice from=bob lamport=28 msg="Indeed. Tres cool"
[Server] 2026/10/24 23:44:40.615845 event=SEND to=bob from=bob lamport=28 msg="Indeed. Tres cool"
[Server] 2026/10/24 23:44:40.615847 event=SEND to=mikkel from=bob lamport=28 msg="Indeed. Tres cool"
[Server] 2026/10/24 23:44:46.451360 event=RECEIVE from=mikkel lamport=31 msg="Yaaas "
[Server] 2026/10/24 23:44:46.451387 event=BROADCAST kind=MESSAGE client=mikkel lamport=31
[Server] 2026/10/24 23:44:46.451402 event=SEND to=bob from=mikkel lamport=31 msg="Yaaas "
[Server] 2026/10/24 23:44:46.451413 event=SEND to=mikkel from=mikkel lamport=31 msg="Yaaas "
[Server] 2026/10/24 23:44:46.451416 event=SEND to=alice from=mikkel lamport=31 msg="Yaaas "
[Server] 2026/10/24 23:44:49.268752 event=DISCONNECT client=mikkel lamport=32
[Server] 2026/10/24 23:44:49.268810 event=BROADCAST kind=LEAVE client=mikkel lamport=32
[Server] 2026/10/24 23:44:49.268832 event=SEND to=bob from=Server lamport=32 msg="Participant mikkel left Chit Chat at logical time 32"
[Server] 2026/10/24 23:44:49.268828 event=SEND to=alice from=Server lamport=32 msg="Participant mikkel left Chit Chat at logical time
      32"
[Server] 2026/10/24 23:44:52.406155 event=DISCONNECT client=bob lamport=33
[Server] 2026/10/24 23:44:52.406206 event=BROADCAST kind=LEAVE client=bob lamport=33
[Server] 2026/10/24 23:44:52.406246 event=SEND to=alice from=Server lamport=33 msg="Participant bob left Chit Chat at logical time 33"
[Server] 2026/10/24 23:44:54.939608 event=DISCONNECT client=alice lamport=34
[Server] 2026/10/24 23:44:54.939669 event=BROADCAST kind=LEAVE client=alice lamport=34
[Server] 2026/10/24 23:44:58.841439 event=SHUTDOWN addr=localhost:5050 lamport=35
[Server] 2026/10/24 23:44:58.841896 event=STOPPED
[Server] 2026/10/25 17:32:40.637138 event=STARTUP addr=localhost:5050 lamport=0
[Server] 2026/10/25 17:32:53.806436 event=SHUTDOWN addr=localhost:5050 lamport=1
[Server] 2026/10/25 17:32:53.807446 event=STOPPED
```

client_alice_e43ba9bd-f600-4ebc-a766-f37718063f08.log

```
[Client] 2026/10/24 23:43:28.739022 event=RECEIVE client=alice from=Server lamport=2 clock=3 msg="Participant alice joined Chit Chat
      at logical time 2"
[Client] 2026/10/24 23:43:36.455604 event=RECEIVE client=alice from=Server lamport=3 clock=4 msg="Participant bob joined Chit Chat at
      logical time 3"
[Client] 2026/10/24 23:43:47.344716 event=RECEIVE client=alice from=Server lamport=4 clock=5 msg="Participant mikkel joined Chit Chat
      at logical time 4"
[Client] 2026/10/24 23:44:02.104419 event=SEND client=alice lamport=6 msg="Hello I'm alice!"
[Client] 2026/10/24 23:44:02.105227 event=RECEIVE client=alice from=alice lamport=7 clock=8 msg="Hello I'm alice!"
[Client] 2026/10/24 23:44:09.010701 event=RECEIVE client=alice from=bob lamport=10 clock=11 msg="Hi! I'm bob"
[Client] 2026/10/24 23:44:12.874501 event=RECEIVE client=alice from=mikkel lamport=13 clock=14 msg="I'm Mikkel"
[Client] 2026/10/24 23:44:15.261159 event=RECEIVE client=alice from=mikkel lamport=16 clock=17 msg="Mikkel!!"
[Client] 2026/10/24 23:44:27.296933 event=RECEIVE client=alice from=bob lamport=19 clock=20 msg="Hello Mikkel. Omg it's sooo nice to
      meet youu"
[Client] 2026/10/24 23:44:33.176652 event=SEND client=alice lamport=21 msg="Tihi this is working very well"
[Client] 2026/10/24 23:44:33.177296 event=RECEIVE client=alice from=alice lamport=22 clock=23 msg="Tihi this is working very well"
[Client] 2026/10/24 23:44:35.316606 event=SEND client=alice lamport=24 msg="Very cool"
[Client] 2026/10/24 23:44:35.319083 event=RECEIVE client=alice from=alice lamport=25 clock=26 msg="Very cool"
[Client] 2026/10/24 23:44:40.616013 event=RECEIVE client=alice from=bob lamport=28 clock=29 msg="Indeed. Tres cool"
[Client] 2026/10/24 23:44:46.451570 event=RECEIVE client=alice from=mikkel lamport=31 clock=32 msg="Yaaas "
[Client] 2026/10/24 23:44:49.268973 event=RECEIVE client=alice from=Server lamport=32 clock=33 msg="Participant mikkel left Chit Chat
      at logical time 32"
[Client] 2026/10/24 23:44:52.406475 event=RECEIVE client=alice from=Server lamport=33 clock=34 msg="Participant bob left Chit Chat at
      logical time 33"
[Client] 2026/10/24 23:44:54.938571 event=DISCONNECT client=alice reason="interrupted"
```

client_bob_1f8689b3-b582-448b-9dfa-966ef3d13ec2.log

```
[Client] 2026/10/24 23:43:36.455669 event=RECEIVE client=bob from=Server lamport=3 clock=4 msg="Participant bob joined Chit Chat at
      logical time 3"
[Client] 2026/10/24 23:43:47.344786 event=RECEIVE client=bob from=Server lamport=4 clock=5 msg="Participant mikkel joined Chit Chat at
      logical time 4"
[Client] 2026/10/24 23:44:02.105225 event=RECEIVE client=bob from=alice lamport=7 clock=8 msg="Hello I'm alice!"
[Client] 2026/10/24 23:44:09.009885 event=SEND client=bob lamport=9 msg="Hi! I'm bob"
[Client] 2026/10/24 23:44:09.010672 event=RECEIVE client=bob from=bob lamport=10 clock=11 msg="Hi! I'm bob"
[Client] 2026/10/24 23:44:12.874485 event=RECEIVE client=bob from=mikkel lamport=13 clock=14 msg="I'm Mikkel"
[Client] 2026/10/24 23:44:15.261166 event=RECEIVE client=bob from=mikkel lamport=16 clock=17 msg="Mikkel!!"
[Client] 2026/10/24 23:44:27.296202 event=SEND client=bob lamport=18 msg="Hello Mikkel. Omg it's sooo nice to meet youu"
[Client] 2026/10/24 23:44:27.296784 event=RECEIVE client=bob from=bob lamport=19 clock=20 msg="Hello Mikkel. Omg it's sooo nice to
      meet youu"
[Client] 2026/10/24 23:44:33.177246 event=RECEIVE client=bob from=alice lamport=22 clock=23 msg="Tihi this is working very well"
[Client] 2026/10/24 23:44:35.319080 event=RECEIVE client=bob from=alice lamport=25 clock=26 msg="Very cool"
[Client] 2026/10/24 23:44:40.615427 event=SEND client=bob lamport=27 msg="Indeed. Tres cool"
[Client] 2026/10/24 23:44:40.616095 event=RECEIVE client=bob from=bob lamport=28 clock=29 msg="Indeed. Tres cool"
[Client] 2026/10/24 23:44:46.451569 event=RECEIVE client=bob from=mikkel lamport=31 clock=32 msg="Yaaas "
[Client] 2026/10/24 23:44:49.269009 event=RECEIVE client=bob from=Server lamport=32 clock=33 msg="Participant mikkel left Chit Chat at
      logical time 32"
[Client] 2026/10/24 23:44:52.403662 event=DISCONNECT client=bob reason="interrupted"
```

client_mikkel_71984236-588c-4cb3-bccf-ef0532da3d9e.log

```
[Client] 2026/10/24 23:43:47.344823 event=RECEIVE client=mikkel from=Server lamport=4 clock=5 msg="Participant mikkel joined Chit Chat
      at logical time 4"
[Client] 2026/10/24 23:44:02.105355 event=RECEIVE client=mikkel from=alice lamport=7 clock=8 msg="Hello I'm alice!"
[Client] 2026/10/24 23:44:09.010756 event=RECEIVE client=mikkel from=bob lamport=10 clock=11 msg="Hi! I'm bob"
[Client] 2026/10/24 23:44:12.873671 event=SEND client=mikkel lamport=12 msg="I'm Mikkel"
[Client] 2026/10/24 23:44:12.874698 event=RECEIVE client=mikkel from=mikkel lamport=13 clock=14 msg="I'm Mikkel"
[Client] 2026/10/24 23:44:15.260351 event=SEND client=mikkel lamport=15 msg="Mikkel!!"
[Client] 2026/10/24 23:44:15.261115 event=RECEIVE client=mikkel from=mikkel lamport=16 clock=17 msg="Mikkel!!"
[Client] 2026/10/24 23:44:27.296905 event=RECEIVE client=mikkel from=bob lamport=19 clock=20 msg="Hello Mikkel. Omg it's sooo nice to
      meet youu"
[Client] 2026/10/24 23:44:33.177230 event=RECEIVE client=mikkel from=alice lamport=22 clock=23 msg="Tihi this is working very well"
[Client] 2026/10/24 23:44:35.319087 event=RECEIVE client=mikkel from=alice lamport=25 clock=26 msg="Very cool"
[Client] 2026/10/24 23:44:40.615966 event=RECEIVE client=mikkel from=bob lamport=28 clock=29 msg="Indeed. Tres cool"
[Client] 2026/10/24 23:44:46.451029 event=SEND client=mikkel lamport=30 msg="Yaaas "
[Client] 2026/10/24 23:44:46.451581 event=RECEIVE client=mikkel from=mikkel lamport=31 clock=32 msg="Yaaas "
[Client] 2026/10/24 23:44:49.267634 event=DISCONNECT client=mikkel reason="interrupted"
```